

Bases and Integral Relations for the k -Kernel of Euler's Totient

Exact finite-level ranks and consequences for Erdős Problem #249

WILL COOK*

July 2026; revised 16 September 2026

ABSTRACT

For integers $k \geq 2$ and $e \geq 1$, we give a basis of the totient sections $n \mapsto \varphi(k^j n + r)$ through level e . Their rational rank is $k^e + 1$. Explicit scalar reductions give unique integral coordinates and a basis of all integral relations. The independence argument uses CRT–Dirichlet separation, an earlier method of Yazdani credited to Shallit; Martin's theorem already implies the requisite affine independence, and Coons previously proved non- k -regularity. The contribution here is the finite-level normal form, not those qualitative obstructions. We also prove irrationality of $\sum_{n \geq 1} (\varphi(n) \bmod m) 2^{-n}$ for every $m \geq 3$ and classify rational dyadic residue observables. For $S = \sum_{n \geq 1} \varphi(n) 2^{-n}$, we prove a seed-to-ray amplification and record an equivalent central-residue condition. These results do not establish irrationality of S . Ordinary proofs are given in the note; formal counterparts and their precise scope are identified separately.

1 A basis and all its relations

Set $\varphi(0) = 0$. For $k \geq 2$, the k -kernel consists of the sequences $n \mapsto \varphi(k^j n + r)$ with $j \geq 0$ and $0 \leq r < k^j$. Coons proved that φ is not k -regular for any $k \geq 2$ [2, Thm. 3.2], and Martin's Theorem 1 implies that the functions $n \mapsto \varphi(a_i n + b_i)$ are linearly independent along pairwise nonproportional affine forms with integer coefficients and positive slopes [4, Thm. 1]. The following theorem gives the exact rank, unique integral coordinates, and, in Corollary 1.3, all integral relations at every finite level. Section 1.1 gives a direct proof of the independence step, with the antecedent of its separation method credited there. No priority claim is made for affine independence.

Theorem 1.1 (exact rank of every totient k -kernel truncation). *Let $k \geq 2$ and $e \geq 1$ be integers, write $F_{j,r}^{(k)}(n) = \varphi(k^j n + r)$, and put*

$$V_{k,e} = \text{span}_{\mathbb{Q}} \{ F_{j,r}^{(k)} : 0 \leq j \leq e, 0 \leq r < k^j \}.$$

Then $\dim_{\mathbb{Q}} V_{k,e} = k^e + 1$, and

$$\mathcal{B}_{k,e} = \{ F_{0,0}^{(k)}, F_{1,0}^{(k)} \} \cup \{ F_{j,r}^{(k)} : 1 \leq j \leq e, 1 \leq r < k^j, k \nmid r \}$$

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

is a basis. Every omitted section reduces to a basis element by an explicit scalar: $F_{j,0}^{(k)} = k^{j-1}F_{1,0}^{(k)}$ for $j \geq 1$, and if $r = k^t u$ with $t = \max\{s : k^s \mid r\} \geq 1$ and $k \nmid u$, then

$$F_{j,r}^{(k)} = C_k(t, u) F_{j-t, u}^{(k)}, \quad C_k(t, u) = k^t \prod_{\substack{p \mid k \\ p \nmid u}} \left(1 - \frac{1}{p}\right).$$

1. [UNCONDITIONAL PROGRESS] **Exact finite-level rank.** For integers $k \geq 2$ and $e \geq 1$, Theorem 1.1 proves ordinarily that the totient k -kernel truncation has rank $k^e + 1$. The dyadic case is also the [exact finite-truncation rank](#); the reusable all-base Lean wrapper is [the same count from an independent family](#), and retains linear independence as an explicit input. That input is discharged by the independent CRT–Dirichlet proof in `AllBaseTotientKernel.lean`; the [complete truncation rank](#) therefore has no independence hypothesis. Martin supplies the broader external antecedent, rather than an axiom of this Lean proof.

The retained condition is $k \nmid r$. At $k = 6$, for example, the residue $r = 2$ is retained. At base 2 and level 3, the fifteen sections have nine coordinates; the omitted positive-residue sections satisfy

$$\varphi(8n + 2) = \varphi(4n + 1), \quad \varphi(8n + 4) = 2\varphi(2n + 1), \quad \varphi(8n + 6) = \varphi(4n + 3).$$

1.1 The evaluation matrix

The two ingredients are scalar reduction and arithmetic independence. The reductions explain which sections must be dependent. The difficulty is to exclude every additional relation. We do this one coefficient at a time: an evaluation will annihilate all the other columns modulo a prime while leaving the chosen column nonzero. Thus no simultaneous primality assertion for several affine forms is needed. The local formula for φ gives the displayed reductions. For a prime $p \mid k$ that does not divide u , the argument $k^{j-t}n + u$ is a p -adic unit, so multiplication by k^t contributes its power of p and one factor $1 - 1/p$. When $p \mid u$, the argument already contains p , and only its power contributes. This gives $C_k(t, u)$.

The separation method has an earlier source: Yazdani’s proof of Theorem 2, explicitly credited there to Shallit, uses the Chinese remainder theorem and Dirichlet’s theorem to distinguish kernel sections modulo a fixed modulus [10, Thm. 2 and proof, pp. 652–653]. For a later account see Allouche, Shallit and Yassawi [11, Thm. 3 and Rem. 7]. Here one separates a chosen column from every other column of a finite evaluation matrix. Nonsingularity, not merely distinctness of infinitely many sections, gives linear independence.

For independence, consider finitely many forms $L_i(n) = a_i n + b_i$ with positive slopes and $a_i b_j \neq a_j b_i$ for $i \neq j$. A finite shift makes the intercepts positive. Write $L_i = g_i A_i$, where A_i is primitive. Choose an odd prime ℓ avoiding the slopes, all $g_i \varphi(g_i)$, and the nonzero cross determinants. In row i , choose distinct primes $q_{ij} \equiv 1 \pmod{\ell}$ for $j \neq i$, outside the same finite exceptional set, and impose

$$L_j(n) \equiv 0 \pmod{q_{ij}}, \quad A_i(n) \equiv 2 \pmod{\ell}.$$

The Chinese remainder theorem combines these congruences. The resulting progression for $A_i(n)$ is reduced: a prime q_{ij} dividing $A_i(n)$ would also divide the nonzero cross determinant; ℓ is excluded by the chosen residue; and primitivity excludes the primes of the slope. Dirichlet's theorem supplies arbitrarily large primes $p = A_i(n)$ with $p \nmid g_i$. Only the selected affine form is made prime in this row. The diagonal value is $\varphi(g_i)(p-1) \not\equiv 0 \pmod{\ell}$. Each off-diagonal value is divisible by ℓ , since $q_{ij}-1$ divides the corresponding totient. One evaluation per row gives

$$(\varphi(L_j(n_i)))_{i,j} \equiv \text{diag}(u_1, \dots, u_s) \pmod{\ell}, \quad u_i \neq 0.$$

The determinant is nonzero over $\mathbb{Q}$.

Apply this construction after restricting a relation among the retained sections to $n = km + 1$. The positive-residue sections give pairwise nonproportional forms

$$k^{j+1}m + (k^j + r), \quad k \nmid r,$$

and the remaining form is $km + 1$. Equality between two affine fractions would make the residue at the larger level divisible by k . The matrix therefore eliminates every positive-residue coefficient. If A, B are the two zero-residue coefficients, the restriction gives $A + B\varphi(k) = 0$. Evaluation at $n = k$ gives $A + Bk = 0$, since $\varphi(k^2) = k\varphi(k)$. As $\varphi(k) < k$, both coefficients vanish. The reductions give spanning and the count is

$$2 + \sum_{j=1}^e (k^j - k^{j-1}) = k^e + 1.$$

This proves Theorem 1.1. The scalar reduction at level $j - t$, with the Euler product written out, is [maximal power reduction](#).

Corollary 1.2 (Dyadic basis). *The family*

$$\{\varphi_{0,0}, \varphi_{1,0}\} \cup \{\varphi_{j,r} : j \geq 1, 0 < r < 2^j, r \text{ odd}\}, \quad \varphi_{j,r}(n) = \varphi(2^j n + r),$$

is a basis for the rational span of the full dyadic kernel. Every rational relation is generated by the scalar reductions. The complete level-zero truncation has dimension one.

Proof. Every finite subfamily lies in a sufficiently high truncation. The theorem gives its independence; the reductions give equality of the full spans. At level zero the only channel is $\varphi(n)$, which is nonzero since $\varphi(1) = 1$, so its span has dimension one. This is an ordinary proof; it makes no claim about an additional formal declaration. $\square$

Corollary 1.3 (Integral normal form). *Let $k \geq 2$ and $e \geq 1$. The canonical family is a $\mathbb{Z}$ -basis of the module generated by the sections through level e . For each omitted channel F_i , write its canonical reduction as $F_i = a_i F_{j(i)}$, where $j(i)$ is retained and a_i is a nonnegative integer. In the free abelian group on all the channels, the vectors*

$$R_i = E_i - a_i E_{j(i)} \quad (i \text{ omitted})$$

form a $\mathbb{Z}$ -basis of the kernel of evaluation $E_i \mapsto F_i$. In particular, every integral relation has a unique integral expression in these elementary relations, and their rank is $\sum_{j=1}^{e-1} k^j$.

Proof. The denominator $\prod_{p|k, p \nmid u} p$ in $C_k(t, u)$ divides k , so the successive channel reductions have integer scalars. They give integral coordinates in the retained family; its rational independence makes these coordinates unique over $\mathbb{Z}$ as well.

Now let $x = \sum_i x_i E_i$ evaluate to zero. For each omitted index i , subtract $x_i R_i$. The result is supported on retained indices and still evaluates to zero, so independence forces it to vanish. Thus the R_i generate the integral relation module. Conversely, the coefficient at an omitted index i in $\sum_h c_h R_h$ is exactly c_i : each relation has coefficient one at its own omitted coordinate and zero at all the others. This proves independence and uniqueness without dividing by any integer. There are

$$(1 + k + \dots + k^e) - (k^e + 1) = \sum_{j=1}^{e-1} k^j$$

omitted channels. The formal statement combines the integral coordinate basis, these two-term relation vectors, and their exact rank. This complete integral normal form is [kernel-checked in Lean](#). $\square$

For example, at base 2 and depth 2, retain $F_{0,0}, F_{1,0}, F_{1,1}, F_{2,1}, F_{2,3}$. The two omitted channels satisfy $F_{2,0} = 2F_{1,0}$ and $F_{2,2} = F_{1,1}$, so

$$E_{2,0} - 2E_{1,0}, \quad E_{2,2} - E_{1,1}$$

are an integral basis of all relations among the seven channels. The rank calculation alone would give two; the basis also identifies every relation and its unique coefficients.

The module here is the one generated by the channels. It is not saturated in the group of all integer-valued sequences: at base 2 and $e \geq 2$, $\varphi(4n+3)/2$ is integer-valued but has canonical coefficient $1/2$.

The totient is not k -regular, in the sense of Allouche and Shallit [1], for any $k \geq 2$: Coons proved that the $\mathbb{Z}$ -module generated by its k -kernel is not finitely generated, by counting the poles of $\zeta(s-1)/\zeta(s)$ in the strip $0 \leq \operatorname{Re} s \leq 1$ [2, Cor. 3.1, Thm. 3.2]. For integer-valued sequences this is equivalent to infinite dimension of the rational span: if the span has dimension d , some d evaluations map it injectively to $\mathbb{Q}^d$ and the module into $\mathbb{Z}^d$ (compare the evaluation argument in [3, App. A]). Theorem 1.1 therefore recovers Coons's theorem by a different route and gives the dimension at every finite level.

The generating-function context is stronger than non-regularity alone. For multiplicative coefficients in characteristic zero, Bell and Smertnig prove that k -Mahler implies k -regular [12, Thm. 1.3]; their totient example therefore rules out a k -Mahler equation for $F_\varphi(z) = \sum_{n \geq 1} \varphi(n)z^n$ in every base $k \geq 2$. Bell, Bruin and Coons give the corresponding algebraic and D -finite classification framework [13, Thms. 1.5–1.6]. These are restrictions on functions. Neither classification by itself determines the arithmetic nature of $F_\varphi(1/2)$.

For a k -regular sequence the difference between a basis of the kernel span and a redundant spanning family is material: matrices associated to a basis compute the growth exponent through their joint spectral radius, whereas a spanning family gives only an

upper bound [3, Thm. 1, Prop. 4, Cor. 7]. That theorem does not apply to φ , but the section maps $g(n) \mapsto g(kn+i)$ still have an explicit form here. Each sends $F_{j,r}$ to $F_{j+1,k^j i+r}$, so it carries $V_{k,e}$ into $V_{k,e+1}$, and no level is invariant under all of them because the dimensions increase. In the canonical bases, a retained section with $j \geq 1$ goes to a retained section, while $F_{0,0}$ and $F_{1,0}$ go to $F_{1,i}$ and to $F_{2,ki}$, a positive integer multiple of $F_{1,i}$. Every element of $\mathcal{B}_{k,e}$ is therefore sent to a positive integer multiple of an element of $\mathcal{B}_{k,e+1}$.

1.2 A reusable consequence

Corollary 1.4 (Freezing periodic coefficients). *If L_i are pairwise nonproportional positive-slope affine forms and w_i are rational-valued periodic sequences, then*

$$\sum_i w_i(n) \varphi(L_i(n)) = 0 \quad \text{for all sufficiently large } n \quad \implies \quad w_i(n) = 0 \quad \text{for every } i, n.$$

Proof. Choose a common period Q . On each progression $n = a + Qt$, the coefficients are constant and the cross determinants become $Q(a_i b_j - a_j b_i) \neq 0$. Shift beyond the threshold and apply the affine independence just proved. This is an ordinary proof of the periodic-coefficient statement; no additional formal declaration is asserted here. $\square$

Applied on each progression, Martin's Theorem 1 gives the same conclusion [4, Thm. 1]. The same argument preserves the basis after deletion of a finite prefix. For the two zero channels, large $n \equiv 1 \pmod{k}$ and large powers of k replace the two evaluations used above.

2 Bounded residues and rationality

There is an earlier coefficient-level obstruction: for each $m \geq 3$, $\varphi(n) \bmod m$ is not automatic in any integer base $k \geq 2$ [10, Cor. 4, p. 654]; see also [11, Thm. 3 and Ex. 4]. This does not prove irrationality of the dyadic sum. When $m > 2$ the raw residues are not binary digits, and carrying changes the digit sequence. The residue theorem below instead forces neighbouring coefficients to vanish while retaining one central value, then uses a bounded-tail argument. Erick Wong posted a proof that $\sum_{n \geq 1} (\varphi(n) \bmod k) k^{-n}$ is irrational for every integer $k > 2$, with the base equal to the modulus, as an answer on Mathematics Stack Exchange on 29 March 2015 [5]. Theorem 2.1 fixes the base at 2 for every modulus $m \geq 3$ and classifies the dyadic residue observables.

Theorem 2.1. *For every $m \geq 3$,*

$$A_m = \sum_{n \geq 1} \frac{\varphi(n) \bmod m}{2^n} \notin \mathbb{Q}.$$

For $k \geq 1$ and $f : \mathbb{Z}/2^k\mathbb{Z} \rightarrow \mathbb{Q}$, the series $\sum_{n \geq 1} f(\varphi(n) \bmod 2^k) 2^{-n}$ is rational exactly when f is constant on the even residue classes. If that constant is c , its value is $3f(1)/4 + c/4$.

Lemma 2.2 (A bounded isolated pulse). *Let $a_n \in \mathbb{Z}$ satisfy $|a_n| \leq C$. Suppose that for arbitrarily large L there is $N > L$ such that $a_N \neq 0$ and $a_{N+t} = 0$ for $0 < |t| \leq L$. Then $\sum_{n \geq 1} a_n 2^{-n}$ is irrational.*

Proof. If the sum has denominator q , every scaled tail $T_j = \sum_{t \geq 1} a_{j+t} 2^{-t}$ belongs to $q^{-1} \mathbb{Z}$. Choose L so large that $C2^{-L} < 1/q$. The right zero block gives $|T_N| \leq C2^{-L} < 1/q$, hence $T_N = 0$. The left zero block now gives $T_{N-L-1} = a_N 2^{-L-1}$, a nonzero element of $q^{-1} \mathbb{Z}$ with absolute value less than $1/q$, a contradiction. The statement is proved ordinarily above. The separate formal counterpart is in [wcook04/plectis-erdos-lean](#) at commit 52f29ad1 states this lemma as `irrational_dyadicValue_of_pulses` in `PalomarCorpus/E249/Challenge.lean` and proves it. That repository's Linux replay of Palomar's Comparator stage accepted the entry with the Lean kernel and with NanoDa ([run 34782407633](#)). $\square$

Proof of Theorem 2.1. Fix a unit $s \pmod{m}$ and a block length L . For each nonzero $t \in [-L, L]$, choose a distinct prime $q_t \equiv 1 \pmod{m}$ larger than L and outside the prime divisors of m . Prescribe

$$p \equiv s \pmod{m}, \quad p \equiv -t \pmod{q_t}.$$

This is a reduced progression, because each $q_t > |t|$. Dirichlet supplies arbitrarily large primes p in it. Then $m \mid \varphi(p+t)$ for every $0 < |t| \leq L$, whereas $\varphi(p) \equiv s-1 \pmod{m}$.

Subtract $f(0)$ from the observable and clear its denominators. Whenever $f(s-1) \neq f(0)$, the preceding construction gives arbitrarily long bounded isolated pulses, so Lemma 2.2 applies. For the least-residue map choose $s = -1$: its central residue is $m-2 \neq 0$ when $m \geq 3$. For $m = 2^k$, every even residue r has $r+1$ a unit, so a non-constant restriction to the even residues supplies such a pulse. Conversely, $\varphi(n)$ is even for every $n \geq 3$. Constancy on the even classes therefore makes all terms after $n = 2$ constant, giving the stated rational value. This converse requires no recurrent-support theorem. The three clauses together are ordinary in this note (that `PaperCompleteR7` file is absent). $\square$

The bounded-pulse proof fixes its coefficient bound before choosing the block. For S , the termwise condition $2^L \mid \varphi(N+L)$, with $N+L > 1$, forces $2^L \leq \varphi(N+L) \leq N+L-1$. Hence the tail envelope $(N+L+2)2^{-L} > 1$ cannot exclude any positive denominator. The following criterion instead measures cancellation in a weighted prefix.

3 The actual series and the remaining residue gap

Write

$$S = \sum_{n \geq 1} \varphi(n) 2^{-n}, \quad R_N = \sum_{j \geq 1} \varphi(N+j) 2^{-j}, \quad \Delta_h(N) = R_{N+h} - R_N.$$

The prefix identity $2^N S = \Phi_N + R_N$, with $\Phi_N = \sum_{n=1}^N \varphi(n) 2^{N-n} \in \mathbb{Z}$, gives

$$\Delta_h(N) \equiv 2^N (2^h - 1) S \pmod{\mathbb{Z}}. \quad (1)$$

In particular a nonintegral tail difference is a precise arithmetic statement about S . The complementary rationality consequence uses the same carry for both kinds of rigidity: the [same-carry rank and modular-period theorem](#) gives rank at least $2^e - 1$ at every

depth and one common eventual period for all dyadic sections modulo a fixed positive multiplier. It supplies necessary structure under rationality, not a contradiction between the two properties.

3.1 One seed removes the full-depth restriction

For $h, N, L \in \mathbb{N}$, let

$$D_{h,N,L} = \sum_{j=0}^{L-1} (\varphi(N+h+1+j) - \varphi(N+1+j))2^{L-1-j}, \quad B = N+h+L+2.$$

Write $K(h, N, L)$ for

$$B < D_{h,N,L} \bmod 2^L < 2^L - B.$$

The exact truncation and $\varphi(n) \leq n$ give

$$|2^L \Delta_h(N) - D_{h,N,L}| \leq B. \quad (2)$$

Thus some L satisfies $K(h, N, L)$ exactly when $\Delta_h(N) \notin \mathbb{Z}$: necessity follows from the strict inequalities; sufficiency follows by choosing L with $2^L \|\Delta_h(N)\|_{\mathbb{R}/\mathbb{Z}} > 2B$.

Theorem 3.1 (Full-depth amplification). *Fix $d \geq 1$ and $N \geq 0$. If $\Delta_d(N) \notin \mathbb{Z}$, then every sufficiently late pair $\{t, t+1\}$ contains an m such that $K(md, N, md)$ holds. Consequently*

$$\exists t \geq 1 : K(td, N, td) \iff \Delta_d(N) \notin \mathbb{Z}.$$

Requiring this for every $d \geq 1, N \geq 0$ is equivalent to $S \notin \mathbb{Q}$.

The seed-to-ray step is [eventual two-syndetic amplification](#); it is equivalent to irrationality as [global full-depth equivalence](#).

Proof. Set $F_t = \Delta_{td}(N)$. Equation (1) yields $F_{t+1} \equiv 2^d F_t + F_1 \pmod{\mathbb{Z}}$. If $K(td, N, td)$ fails, (2) gives

$$\|F_t\|_{\mathbb{R}/\mathbb{Z}} \leq \epsilon_t, \quad \epsilon_t = 2(N + 2td + 2)2^{-td}.$$

Two adjacent failures would imply $\|F_1\| \leq 2^d \epsilon_t + \epsilon_{t+1}$, impossible for all sufficiently large t because the left side is positive and the right side tends to zero. Conversely, if F_1 is integral, then every F_t is integral by the same recurrence, so no certificate exists. Finally, (1) is nonintegral for every d, N when S is irrational. If $S = a/(2^c v)$ with v odd, take $N = c$ and $d = \varphi(v)$ to make it integral. $\square$

For instance $D_{1,12,16} = -143140$, with residue 53468 modulo 2^{16} and $B = 31$, supplies a seed. The amplifier produces infinitely many full-depth certificates from this finite fact; it does not provide seeds for all d, N .

3.2 A single canonical endpoint

For $c \geq 0$, odd $v \geq 1$ and a positive multiple H of $\varphi(v)$, put

$$M = \frac{2^H - 1}{v}, \quad B_{H,c} = \sum_{j=0}^{H-1} \varphi(c+1+j)2^{H-1-j}, \quad K = c + H + 1.$$

Euler's theorem makes M integral.

Equivalent formulation 3.2 (Canonical Mersenne gap). The series S is irrational if and only if, for every $c \geq 0$ and odd $v \geq 1$, some positive multiple H of $\varphi(v)$ satisfies

$$K < (-B_{H,c}) \bmod M < M - K. \quad (3)$$

Proof. The identity

$$B_{H,c} = 2^H R_c - R_{c+H} = M(vR_c) - \Delta_H(c)$$

is exact. For $H \geq 1$, $|\Delta_H(c)| < c + H + 1 = K$. Indeed $R_N \leq N + 1$ for $N \geq 1$ and $0 < R_0 \leq 3/2$; the strict inequalities follow by subtracting a positive tail. If $S = a/(2^c v)$, then $vR_c \in \mathbb{Z}$ and $\Delta_H(c) \in \mathbb{Z}$. The residue of $-B_{H,c}$ is therefore within distance less than K of $0 \pmod{M}$, so (3) fails.

If S is irrational, fix c, v . Then vR_c is nonintegral. As H tends to infinity through positive multiples of $\varphi(v)$, $\Delta_H(c)/M \rightarrow 0$ and $K/M \rightarrow 0$. Hence the fractional part of $-B_{H,c}/M = -vR_c + \Delta_H(c)/M$ stays away from both endpoints of $[0, 1]$, which gives (3) for all sufficiently large such H . $\square$

This proof identifies the equivalent missing quantifier. It does not supply H arithmetically without assuming the desired irrationality. The equivalence, with the modulus written as the natural quotient $(2^H - 1)/v$ and v odd, is ordinary in this note (that PaperCompleteR7 file is absent). The separate public Lean release wcook04/plectis-erdos-lean at commit 52f29ad1 states the equivalence in this form, with $vM = 2^H - 1$, as fullMersenneCanonicalBasepointResidueGapSupply_iff_irrational in PalomarCorpus/E249/Challenge.lean and proves it; the replay receipt is cited at Lemma 2.2. The finite residues also satisfy

$$\rho_{H,N+1,M} = (2\rho_{H,N,M} - \varphi(N+H+1) + \varphi(N+1)) \bmod M, \quad \rho_{H,N,M} = (-B_{H,N}) \bmod M,$$

which is an exact recurrence for searches directed at (3).

4 Information that does not force the gap

1. [EXACT REFORMULATION] **Actual LCM orbit frontier.** Let $H_a = \text{lcm}(1, \dots, 2^a)$ and $R_a = \text{totientTail}(2H_a) - \text{totientTail}(H_a)$. Then S is irrational if and only if, beyond every threshold a_0 , some $a \geq a_0$ has $R_a \notin \mathbb{Z}$. This is the [exact cofinal nonintegrality criterion](#) for the actual LCM diagonal, not a quantitative separation estimate and not a proof that such indices occur; the [forward implication](#) is recorded separately.

2. [UNCONDITIONAL PROGRESS] **Positive LCM corridor and top-edge boundary.** For $a \geq 8$ and $J + (a + 6) < 2 \cdot 2^a$, the true translated orbit difference $\text{totientTail}(2H_a + J) - \text{totientTail}(H_a + J)$ is strictly positive. If it is represented by an integer and the dyadic modulus has room for the directed endpoint strip, the carry survivor is strictly negative and the discrepancy residue is exactly the top-edge representative $2^K - \text{carry}_K$, lying strictly between the strip's lower edge and 2^K . Thus the positive corridor eliminates the nonnegative true survivor but does not itself produce a central-band contradiction; an independent arithmetic exclusion of the top boundary is still required. *Checked: positive corridor, negative survivor, top-edge residue.*
3. [UNCONDITIONAL PROGRESS] **LCM factor-ideal shift-algebra no-go.** For every $t \geq 3$, with $H = \text{lcm}(1, \dots, t)$, there is a nonzero synthetic dyadic coboundary whose forcing letters reproduce the actual totient differences at $t - 2$ prescribed indices and lie in the ideal generated by $\varphi(H)$. Every finite integer shift polynomial preserves the exact coboundary cancellation, the H -factor ideal and all lower factor ideals, together with uniform state and letter bounds. A sparse-anchor variant also preserves every prescribed whole-ray anchor. Hence factor ideals, finite shift-algebra closure and these bounds cannot by themselves force a contradiction. The witness is synthetic: its letters are not claimed to be actual totient differences, nonlinear combinations are outside the theorem, and no cofinal family of certificates follows. *Checked: finite shift-algebra no-go.*

A rational control, proved in `ParityPerturbedRationalControl.lean`, has $c(n) = \varphi(n)$ for odd n , $|c(n) - \varphi(n)| \leq 2$ for even n , $0 \leq c(n) \leq n$, and

$$\sum_{n \geq 1} c(n) 2^{-n} = \frac{5}{4}.$$

It has an integral tempered carry with dyadic rank at least $2^e - 1$ at every level. Thus the same lower bound for a hypothetical totient carry is compatible with rationality. For one fixed prime p and integer-valued g , the exact laws $g(pn) = pg(n)$ when $p \mid n$ and $g(pn) = (p-1)g(n)$ otherwise, together with $g - \varphi = o(n)$, force $g = \varphi$: a nonzero defect keeps a nonzero ratio to the argument along $p^j n$. The control fails these laws; its full coefficient rank is asserted only at measured finite depths.

A different obstruction concerns rational approximation. Put

$$\Theta_2 = \sum_{d \geq 1} \frac{\mu(d)}{(2^d - 1)^2} = S - \frac{1}{2}, \quad Q(e, Y) = \frac{(\sum_{d=1}^Y \mu(d)/(2^d - 1)^{e+2})^2}{\sum_{d=1}^Y \mu(d)/(2^d - 1)^{2e+2}}.$$

The divisor-convolution identity $\varphi = \mu * \text{Id}$ gives the displayed value of Θ_2 . Amiram Eldar posted the corresponding formula $S = \frac{1}{2} + \sum_{d \geq 1} \mu(d)/(2^d - 1)^2$, together with its coprimality interpretation, to OEIS A256936 on 15 March 2026 [6], and Steve Fan posted the formula on the problem's forum thread on 16 May 2026 [7]. The following exact extremum is stated in the long record as Theorem `thm:rankonefloor`; its proof is [the sharp-floor argument](#) in the linked source.

Theorem 4.1 (Positive rank-one floor). *For $e \geq 1$ and $Y \geq 4$, the denominator of $Q(e, Y)$ is positive, and the unique minimiser is $(e, Y) = (1, 5)$. Every admissible quotient and every nonempty finite positive weighted average of such quotients satisfies*

$$Q - \Theta_2 > \frac{21}{320}.$$

The uniform bound $Q - \Theta_2 > 1/15$ is false already at $(1, 5)$.

The unique minimiser and the [uniform 21/320 floor](#) are in the linked source; the $1/15$ bound [already fails at \$\(1, 5\)\$](#) . This theorem treats averages after the quotients are formed. Signed coefficients, coupling before the quotient, and higher-rank kernels require separate arguments. For any proposed rational approximants p_j/q_j in lowest terms, a sufficient arithmetic target is

$$0 < q_j |\Theta_2 - p_j/q_j| \rightarrow 0.$$

Ordinary convergence and Hankel nonvanishing do not contain this denominator estimate. The canonical gap (3) remains the equivalent endpoint; the approximation estimate is one stronger sufficient route.

5 Antecedents and proof record

Allouche and Shallit introduced k -regular sequences [1]. Coons proved that φ is not k -regular [2, Thm. 3.2]; the exact ranks recover this, as explained after Corollary 1.3. Martin’s Theorem 1 makes the ratios of totients on any prescribed finite list of pairwise nonproportional affine forms with integer coefficients and positive slopes arbitrarily separated; this implies their linear independence [4, Thm. 1]. Section 1.1 gives a direct determinant proof tailored to the sections used here. The basis theorem gives the exact finite-level normal form. Section 2 treats the residue series at base two, where Wong’s matched-base answer is credited, and classifies dyadic observables. Erdős and Graham record the irrationality of S as probably hopeless to prove at present [8, p. 61].

The [dyadic finite-level rank \$2^e + 1\$](#) , residue-series irrationality, [full-depth amplification](#), canonical equivalence and the [positive rank-one floor](#) are in the public snapshot. The unconditional all-base exact-rank theorem is also [kernel-checked](#): its CRT–Dirichlet independence proof does not assume Martin’s theorem. [Rank from an independent all-base family](#) remains the reusable conditional factorisation. The [canonical gap](#) is the remaining arithmetic endpoint. The historical source links retain 99f4bf47422a. The integral normal form has its ordinary proof above. The periodic transport lemma is checked in `PeriodicTotientIndependence.lean`; the eventual version follows by the displayed threshold shift. The rational control and the termwise-window and prime-dilation boundaries are proved in `ParityPerturbedRationalControl.lean` and `PrefixValuationAndControlRigidity.lean`, respectively. Finite evaluation ranks and the all-level carry lower bound remain distinct. The long record gives the Lean-checked Farey-window exclusion at window $K = 240$: $S \neq a/q$ for every integer a and every $0 < q \leq Q_0$, where $Q_0 \approx 7.96 \times 10^{34}$ is stated exactly there. It does not imply the universal gap.

The remaining first-harmonic obligation is a four-term pivot budget, not an opening statement of this note. Write

$$\begin{aligned} \text{Re pivotCenteredCorrelation} &\leq \frac{14}{25}X, \\ \|\text{pivotFiberMean}\| &\leq \frac{1}{100}X, \\ \|\text{pivotBad}\| &\leq \frac{1}{100}X, \\ \|\text{pivotNonSupplier}\| &\leq \frac{8}{25}X. \end{aligned}$$

Problem 5.1 (first-harmonic anti-concentration on supplier fibres). For each $h \geq 1$, do there exist $s \geq 1$ and $0 < \eta < 1$ such that, for every X_0 , there are X, L with

$$\max(X_0, 1) \leq X, \quad h \leq L - s, \quad 16(2X + h + L + 2) \leq 2^L,$$

for which those four bounds hold?

The quantities are defined with the [budget condition](#) in the long record. This note does not prove them. Balasubramanian, Giri and Srivastav prove asymptotics for shifted convolutions $\sum_n F(n)G(n-h)$ of divisor convolutions $F = f * 1, G = g * 1$ with decaying f, g , a class that contains $\varphi(n)/n$ [9, Thm. 2.2]; the first bound above needs cancellation against the dyadic phase, which that theorem does not supply.

A Guide to the formal sources

The public snapshot used by this note is recorded by the following source links.

[totient Kernel zero residue](#), [totient Kernel even residue reduce](#), [two pow mul totient series eq](#), [definition](#), [certificate completeness](#), [tail-difference integrality](#), [irrationality iff every tail difference is nonintegral](#), [certified Kill all small](#), [tail diff not int small](#), [is Tempered Binary Orbit](#), [residue recurrence](#), [dominance over the remote condition](#), [exact equivalence](#), [eventual two-syndeticity](#), [pointwise equivalence](#), [global equivalence](#), [cofinal depth deletion](#), [cofinal endpoint](#), [independence](#), [span equality](#), [basis](#), [independence](#), [dimension](#), [the exact finite-truncation rank](#), [infinite-dimensionality directly](#), [unique minimizer](#), [uniform 21/320 floor](#), [1/16 floor](#), [1/15 failure](#), [positive direct sums](#), [rational linear-form obstruction](#), [coordinate specification](#), [injective residues](#), [unique representation](#), [level count](#), [section-index count](#), [index as two plus the odd-residue count](#), [full index count](#), [exclusion](#), [reduced-denominator form](#), [first failure](#), [the barrier](#), [rank transport from 1](#), [the modular-period/rank frontier](#), [forcing vanishing](#), [geometric reduction](#), [identity](#), [sign](#), [primes](#), [prime powers](#), [unboundedness](#), [divisor-count layer](#), [first gcd moment](#), [exact centre decomposition](#), [sharp enclosure](#), [separation criterion](#), [target equivalence](#), [endpoint nonnegativity](#), [foreign-channel identity](#), [five-point loss bound](#), [simultaneous finite family](#), [finite horizon](#), [vanishing block](#), [nonzero block](#), [centrality equivalence](#), [equivalence of the two conditions](#), [conditional irrationality theorem](#), [finite complement bound](#), [exact channel split](#), [conditional target-miss theorem](#), [exact frontier](#), [forward implication](#), [positive corridor](#), [negative survivor](#), [top-edge residue](#), [finite shift-algebra no-go](#), [sparse-anchor no-go](#), [the periodic theorem](#), [no fixed index](#), [the primordial divisibility](#), [pointwise](#), [cofinal at each shift](#), [lcm diagonal](#), [window-separated pairs](#), [the conditional implication](#), [irrational erdos Borwein series](#), [tsum moebius div](#)

two pow sub one eq half, tsum totient div two pow sub one eq two, totient series eq half add moebius mersenne square, totient series eq half add visible coprime pairs, tsum pos coprime inv mersenne eq one, cylinder Mass split, parity Separated Matrix det ne zero, exists separated Minor Certificate totient Affine Odd Family, binary Coeff Series rational iff exists tempered Binary Orbit, farey gap, gap check window 1 240 le 79639646646701375323355774875831053, tsum totient div pow two ne int div of totient gap certificate, gap check window 1 120 le 248672326362367909, tsum visible coprime pairs ne int div of den le 39819823323350687661677887437915526, diagonal split, the $t = 2^4$, length-23 kill, the $t = 2^6$, length-93 kill, actual-orbit non-integrality at both heights, but explicitly stops at that finite prefix, recorded here, by the normalization identity, through the exact threshold lemma, the cofinal separation hypothesis, the conditional endpoint theorem, the staircase is impossible under the stated room hypothesis, terminal-remainder identity, fixed Precision Tropical No Go, asks, a positive adaptive truncation budget, S is irrational, density hypothesis, block-gap hypothesis, density-to-gap implication, exact squaring recurrence, iterate formula, initial-phase normal form, initial-phase equivalence, dyadic-root obstruction, prime-alignment implication, $s = 1$, $s = 2$, first difference, second difference, mixed difference, separable case, fixed-stencil uniqueness, a four-point divisor-layer identity, a centred representative for a residue class modulo an even modulus, Layer, BoundedOrder, FinitePrimeEscape, definition, totient Tail, not irrational of tail diff mem int, list, verification, exists diagonal Kill le 82, CyclotomicAnchoredKillSupply, the checked equivalence, checked conditional support/period no-go, unbounded prime divisors, the checked binary-layer theorem, ApFullDepthEscape, membership, injectivity, image equality, checked counterexample, window First Exp sum eq pivot decomposition, the conditional irrationality theorem, the budget condition, the first-harmonic gap, power-multiplication identity, affine cross-gcd identity, nonzero reduction scalar, all-base residue step, canonical spanning, span equality, canonical all-base rank from independence, all-base truncation rank from independence. At the later public checkpoint 25ef6245d15a, `linearIndependent_allBaseCanonicalFamily`, `finrank_allBaseTotientKernel_eq`, and `finrank_allBaseThroughLevelFamily_eq` provide the unconditional all-base independence and both exact ranks.

References

- [1] J.-P. Allouche and J. Shallit, *The ring of k -regular sequences*, Theoret. Comput. Sci. **98** (1992), no. 2, 163–197, doi:10.1016/0304-3975(92)90001-V.
- [2] M. Coons, *(Non)Automaticity of number theoretic functions*, J. Théor. Nombres Bordeaux **22** (2010), no. 2, 339–352, doi:10.5802/jtnb.718. Theorem 3.2, p. 348.
- [3] M. Coons, *Regular sequences and the joint spectral radius*, Internat. J. Found. Comput. Sci. **28** (2017), no. 2, 135–140, doi:10.1142/S0129054117500095; arXiv:1511.07535v1. Theorem 1, Proposition 4, Corollary 7 and Appendix A are cited with the arXiv v1 numbering.
- [4] G. Martin, *Simultaneous inequalities among values of the Euler phi-function*, arXiv:math/0603053v1, 2006. Theorem 1, pp. 1–2.

- [5] Erick Wong, [answer 1211557](#) to *An infinite sum based on the mod-parity of Euler's totient function*, Mathematics Stack Exchange, 29 March 2015, accessed 15 September 2026.
- [6] A. Eldar, comment and formula added to OEIS A256936 (revisions 28 and 31), 15 March 2026, <https://oeis.org/history?seq=A256936>, accessed 16 September 2026.
- [7] S. Fan, comment on Erdős Problem #249, 16 May 2026, 19:01, [Erdős Problems discussion thread](#), accessed 16 September 2026.
- [8] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, Monographies de L'Enseignement Mathématique **28**, 1980, p. 61.
- [9] R. Balasubramanian, S. Giri and P. Srivastav, *On correlations of certain multiplicative functions*, J. Number Theory **174** (2017), 221–238, doi:[10.1016/j.jnt.2016.10.001](https://doi.org/10.1016/j.jnt.2016.10.001); arXiv:[1511.02221v3](https://arxiv.org/abs/1511.02221v3). Theorem 2.2 is cited with the arXiv v3 pagination, p. 2.
- [10] S. Yazdani, *Multiplicative functions and k -automatic sequences*, J. Théor. Nombres Bordeaux **13** (2001), no. 2, 651–658, https://www.numdam.org/item/JTNB_2001__13_2_651_0/. Theorem 2 and its proof, pp. 652–653; Corollary 4, p. 654.
- [11] J.-P. Allouche, J. Shallit and R. Yassawi, *How to prove that a sequence is not automatic*, Exposition. Math. **40** (2022), no. 1, 1–22, doi:[10.1016/j.exmath.2021.08.001](https://doi.org/10.1016/j.exmath.2021.08.001); arXiv:[2104.13072v1](https://arxiv.org/abs/2104.13072v1). Theorem and example numbering follows the arXiv v1 (2021).
- [12] J. Bell and D. Smertnig, *Mahler series with multiplicative coefficient sequences*, arXiv:[2603.23456v1](https://arxiv.org/abs/2603.23456v1), 24 March 2026, preprint. Theorem 1.3, p. 2; Lemma 3.3, p. 9.
- [13] J. P. Bell, N. Bruin and M. Coons, *Transcendence of generating functions whose coefficients are multiplicative*, Trans. Amer. Math. Soc. **364** (2012), no. 2, 933–959, doi:[10.1090/S0002-9947-2011-05479-6](https://doi.org/10.1090/S0002-9947-2011-05479-6); arXiv:[1003.2221v2](https://arxiv.org/abs/1003.2221v2). Theorems 1.5–1.6 are cited with the arXiv v2 pagination, p. 2.

Companion system context. The [claim and trust boundary](#), [cold-clone route to proof authority](#), and [public contribution protocol](#) are described in sibling papers. Those descriptions do not change the mathematical status of this note.